

Trend DevOps Project Deployment Report

Project: Production Ready React Application Deployment using Docker, Jenkins, GitHub, DockerHub, Terraform and AWS EKS

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1. Project Overview

This project demonstrates a complete DevOps CI/CD pipeline for deploying a React application in a production-ready environment using Docker, Jenkins, GitHub, DockerHub, Terraform, and Kubernetes on AWS EKS.

2. Application Setup

The Trend React application repository was cloned from GitHub and configured to run on port 3000. The production build was containerized using Docker.

3. Docker Implementation

A Dockerfile was created to package the React application inside an Nginx container. Docker images were built locally and tested successfully using Docker Desktop.

4. GitHub Repository

The full codebase including Dockerfile, Jenkinsfile, Kubernetes manifests, Terraform configuration, .gitignore, and .dockerignore files was pushed to GitHub.

5. Jenkins CI/CD Pipeline

Jenkins was configured with required plugins including Git, Docker, Kubernetes, and Pipeline plugins. A declarative Jenkins pipeline was implemented to automate cloning, building Docker images, pushing to DockerHub, and deploying the application.

6. DockerHub Integration

DockerHub credentials were securely configured inside Jenkins. The CI/CD pipeline automatically pushes updated Docker images to DockerHub after every successful build.

7. GitHub Webhook Integration

GitHub webhook integration was configured to automatically trigger Jenkins builds whenever new commits are pushed to the repository.

8. Terraform Infrastructure

Terraform was used to provision AWS infrastructure components including networking, IAM configuration, and EC2 resources for Jenkins deployment.

9. AWS EKS Kubernetes Deployment

An Amazon EKS cluster was created using eksctl. Kubernetes deployment.yaml and service.yaml files were created to deploy the application with a LoadBalancer service.

10. Kubernetes Service and LoadBalancer

The application was exposed publicly using a Kubernetes LoadBalancer service. AWS automatically provisioned the external LoadBalancer endpoint.

11. Monitoring and Future Scope

The project architecture supports future monitoring integration using Prometheus and Grafana for cluster and application health monitoring.

Important Links

- **GitHub Repository:** <https://github.com/prats00017/trend-devops-project>
- **DockerHub Repository:** <https://hub.docker.com/r/prateek0017/trend-devops-app>

Screenshots Section

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\prateek> git clone https://github.com/Vennilavanguvi/Trend.git
Cloning into 'Trend'...
remote: Enumerating objects: 80, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 80 (delta 0), reused 0 (delta 0), pack-reused 79 (from 1)
Receiving objects: 100% (80/80), 8.58 MiB | 6.04 MiB/s, done.
Resolving deltas: 100% (1/1), done.
PS C:\Users\prateek> cd Trend
PS C:\Users\prateek\Trend> dir

Directory: C:\Users\prateek\Trend

Mode                LastWriteTime         Length Name
----                -
d-----          5/22/2026   5:59 PM             dist
-a-----          5/22/2026   5:59 PM          1801 README.md

PS C:\Users\prateek\Trend> |
```

```
PS C:\Users\prateek\Trend> cd dist
PS C:\Users\prateek\Trend\dist> dir

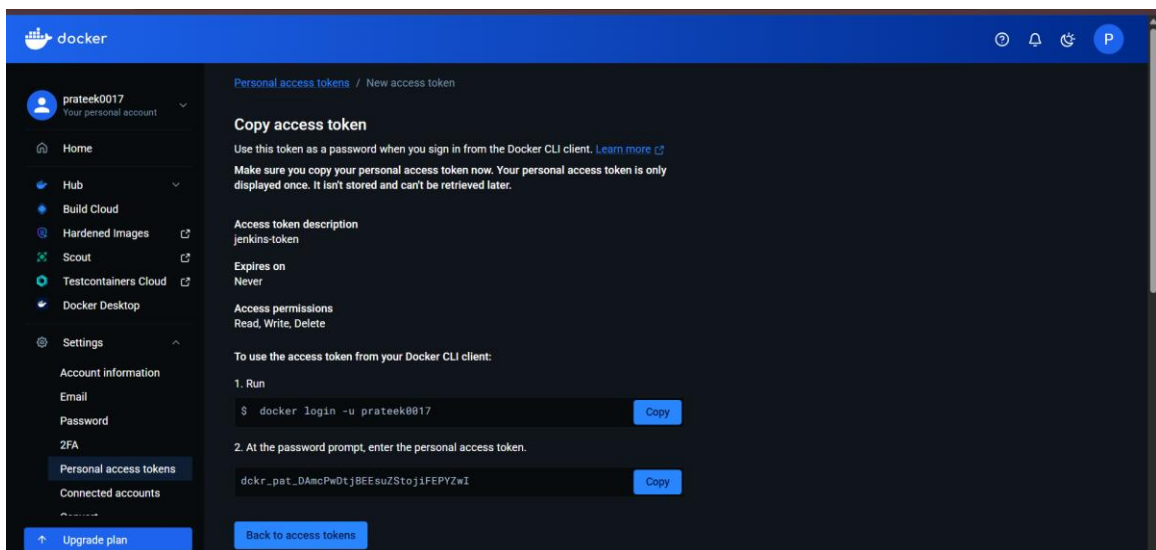
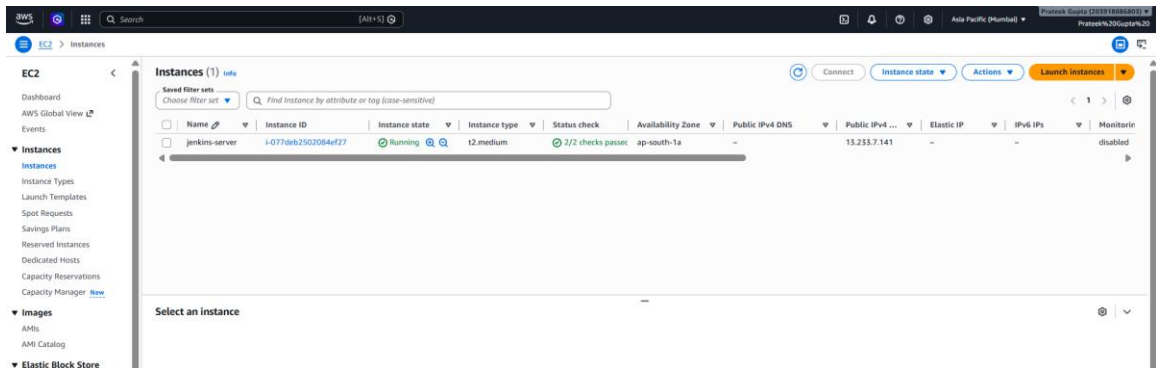
Directory: C:\Users\prateek\Trend\dist

Mode                LastWriteTime         Length Name
----                -
d-----          5/22/2026   5:59 PM             assets
-a-----          5/22/2026   5:59 PM           428 index.html
-a-----          5/22/2026   5:59 PM          1497 vite.svg
```

```

PS C:\Users\prateek\Trend> docker build -t trend-app .
[+] Building 2.6s (8/8) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 140B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [auth] library/nginx:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 55B
=> [internal] load build context
=> => transferring context: 9.24MB
=> CACHED [1/2] FROM docker.io/library/nginx:alpine@sha256:2f07d83bf561b506400dc183b1b2003803e39efbd22451f848ada
=> => resolve docker.io/library/nginx:alpine@sha256:2f07d83bf561b506400dc183b1b2003803e39efbd22451f848adaba14d28
=> [2/2] COPY dist /usr/share/nginx/html
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:2cd9c331827e9765fb6a08b574ec73d2f5f0e756c3b5113b2e9d48bd2d932bd5
=> => exporting config sha256:e1388b209387bac743268c797a1609e48211fd8c8cad29d3a816dabee23b0cbf
=> => exporting attestation manifest sha256:f3b919e9942d211432a5680c6175dcf66b5d49051bc394ea7ef595104ec84bf0
=> => exporting manifest list sha256:157f327f627a33308ce2bdc8e4e69fe09b2345cf48c19463a62fa9680b037ad8
=> => naming to docker.io/library/trend-app:latest
=> => unpacking to docker.io/library/trend-app:latest
PS C:\Users\prateek\Trend>

```



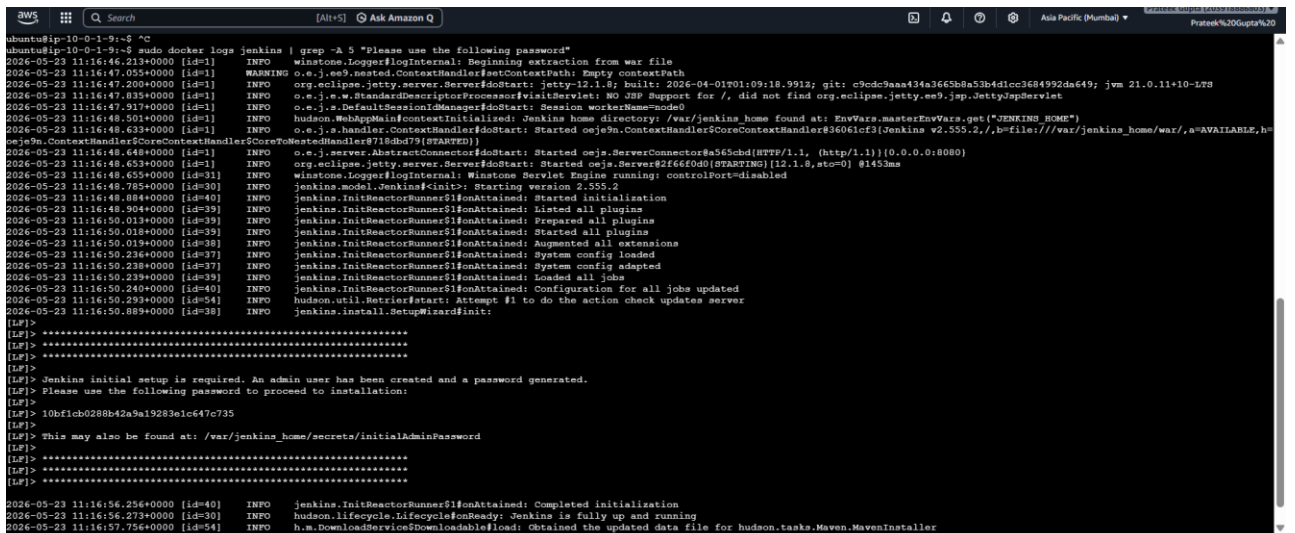
```
aws [Search] [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) Prateek Gupta (203918886803) Prateek%20Gupta%20

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-1-9:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3-24.04.2
ubuntu@ip-10-0-1-9:~$ sudo systemctl start docker
ubuntu@ip-10-0-1-9:~$ sudo systemctl enable docker
ubuntu@ip-10-0-1-9:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-05-23 11:12:16 UTC; 46s ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 57578 (dockerd)
       Tasks: 9
    Memory: 23.0M (peak: 23.4M)
         CPU: 280ms
    CGroup: /system.slice/docker.service
            └─57578 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

May 23 11:12:15 ip-10-0-1-9 dockerd[57578]: time="2026-05-23T11:12:15.827736674Z" level=info msg="Restoring containers: start."
May 23 11:12:15 ip-10-0-1-9 dockerd[57578]: time="2026-05-23T11:12:15.859695656Z" level=info msg="Deleting nftables IPv4 rules" error="exit status 1"
May 23 11:12:15 ip-10-0-1-9 dockerd[57578]: time="2026-05-23T11:12:15.866025176Z" level=info msg="Deleting nftables IPv6 rules" error="exit status 1"
May 23 11:12:16 ip-10-0-1-9 dockerd[57578]: time="2026-05-23T11:12:16.130656072Z" level=info msg="Loading containers: done."
May 23 11:12:16 ip-10-0-1-9 dockerd[57578]: time="2026-05-23T11:12:16.137126390Z" level=info msg="Docker daemon" commit="29.1.3-0ubuntu3-24.04.2" containerd-snapshotter=
May 23 11:12:16 ip-10-0-1-9 dockerd[57578]: time="2026-05-23T11:12:16.137224285Z" level=info msg="Initializing buildkit"
May 23 11:12:16 ip-10-0-1-9 dockerd[57578]: time="2026-05-23T11:12:16.150235637Z" level=info msg="Completed buildkit initialization"
May 23 11:12:16 ip-10-0-1-9 dockerd[57578]: time="2026-05-23T11:12:16.155325284Z" level=info msg="Daemon has completed initialization"
May 23 11:12:16 ip-10-0-1-9 dockerd[57578]: time="2026-05-23T11:12:16.155349001Z" level=info msg="API listen on /run/docker.sock"
May 23 11:12:16 ip-10-0-1-9 systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-22/22 (END)
```

```
*** System restart required ***
Last login: Sat May 23 11:01:20 2026 from 13.233.177.3
ubuntu@ip-10-0-1-9:~$ sudo docker run -d --name jenkins -p 8080:8080 -p 50000:50000 jenkins/jenkins:lts
Unable to find image 'jenkins/jenkins:lts' locally
lts: Pulling from jenkins/jenkins
cebc1e77ceba: Pull complete
c2314d42ccal: Pull complete
307f8152a55e: Pull complete
76a355a69c2e: Pull complete
3bdbd616c406: Pull complete
5834ae4b8774: Pull complete
6b35bb76b8c8: Pull complete
ae218d1cfa6e: Pull complete
801814f62c99: Pull complete
9014a3401c36: Pull complete
4c140080734e: Pull complete
dd00b4ealf64: Pull complete
653a4e57bccf: Download complete
Digest: sha256:01c992ffef29dcf41c7164e8c16285c657c2368c4943dda8d68c93fdf54447d5
Status: Downloaded newer image for jenkins/jenkins:lts
573a46c260da793a856cb44a145d703aa35c66362b5415c9b198e7d45945b797
ubuntu@ip-10-0-1-9:~$
```



← → ↻ 🏠 ⚠ Not secure 13.233.7.141:8080 🔍 ☆ 📄 🗑 📧 ⚙

Getting Started

Create First Admin User

Username

Password

Confirm password

Full name

E-mail address

Jenkins 2.555.2 [Skip and continue as admin](#) [Save and Continue](#)

← → ↻ 🏠 ⚠ Not secure 13.233.7.141:8080 🔍 ☆ 📄 🗑 📧 ⚙

Getting Started

Instance Configuration

Jenkins URL:

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

Jenkins 2.555.2 [Not now](#) [Save and Finish](#)

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

REST API

Jenkins 2.555.2

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

Jenkins

Manage Jenkins

Plugins

Plugins

Search available plugins

Install

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

Install	Name	Released	Health
☒	Pipeline Graph Analysis 254.v0f63a_a_447dca_ Library plugins (for use by other plugins) Provides a REST API to access pipeline and pipeline run data.	3 mo 17 days ago	97
☐	PAM Authentication 1.12 Security Adds Unix Pluggable Authentication Module (PAM) support to Jenkins	1 yr 2 mo ago	90
☐	JavaMail API 1.6.2-11 Library plugins (for use by other plugins) This plugin provides the JavaMail API for other plugins.	1 yr 2 mo ago	94
☐	Command Agent Launcher 134.v025a_5fcf9dea_ Agent Management Allows agents to be launched using a specified command.	1 mo 16 days ago	97
☐	Oracle Java SE Development Kit Installer 83.v417146707a_3d Allows the Oracle Java SE Development Kit (JDK) to be installed via download from Oracle's website.	1 yr 4 mo ago	94
☐	SSH server 3.384.vc89b_5e138cf9 Adds SSH server functionality to Jenkins, exposing CU commands through it.	3 mo 17 days ago	100
☐	Pipeline: REST API 2.41		

Jenkins

trend-app-pipeline

Configure

Configure

General

Triggers

Pipeline

Advanced

10~>stage('HELLO') {
7~steps {
8~echo 'Jenkins Pipeline Working Successfully!'
9~}
10~}
11~
12~stage('Docker Check') {
13~steps {
14~sh 'docker --version'
15~}
16~}

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Advanced

Advanced

Save

Apply

REST APIJenkins 2.555.2

Jenkins

trend-app-pipeline

Status

Changes

Build Now

Configure

Delete Pipeline

Stages

Rename

Pipeline Syntax

trend-app-pipeline

Permalinks

Add description

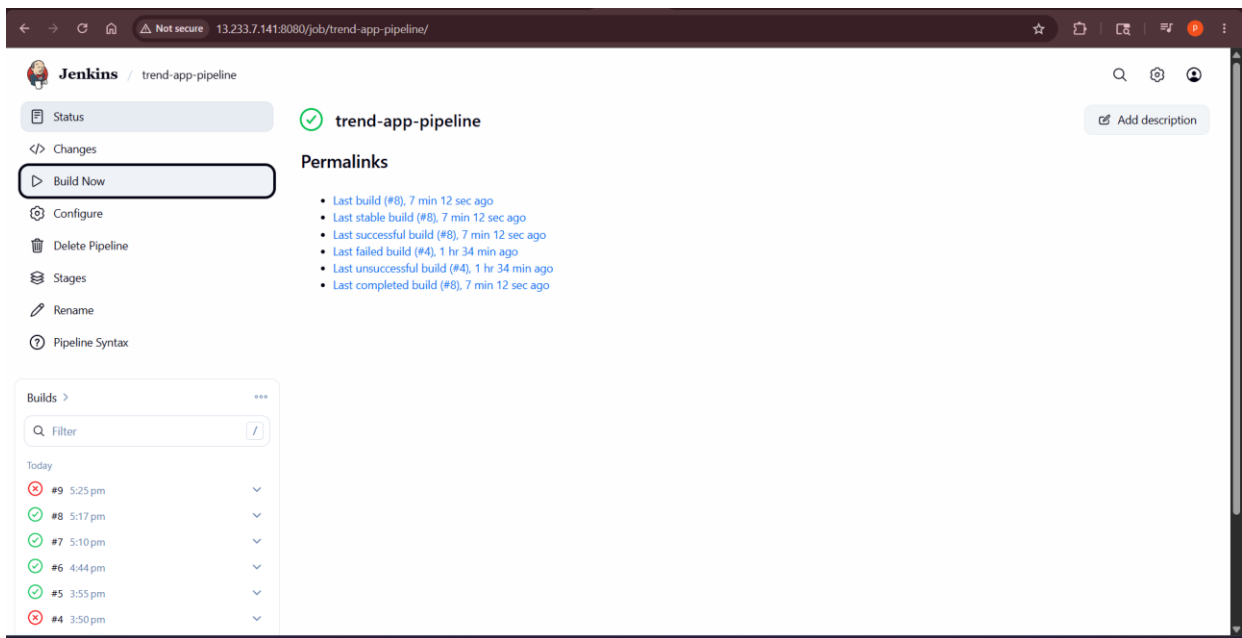
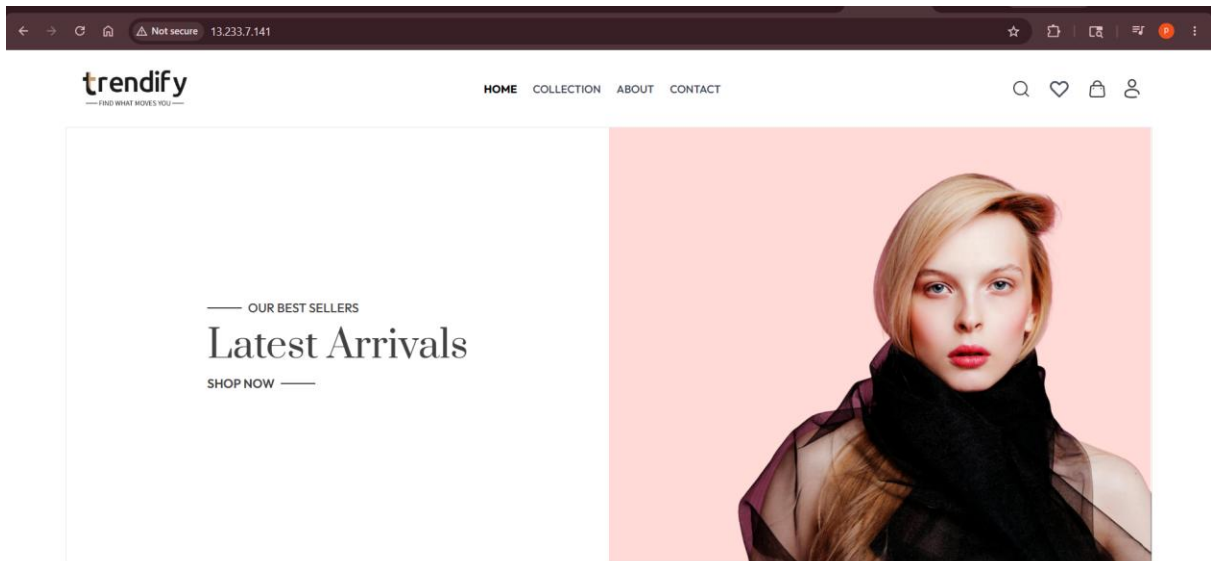
Builds >

TodayTook 3.4 sec

#1 11:53 am

13.233.7.141:8080/job/trend-app-pipeline/1/

REST APIJenkins 2.555.25:23 PM5/23/2026



```
Administrator: Windows PowerShell

2 sequential sub-tasks: {
  2 sequential sub-tasks: {
    1 task: { create addons },
    wait for control plane to become ready,
  },
  create managed nodegroup "trend-nodes",
}

0026-05-23 23:39:23 [ ] building cluster stack "eksctl-trend-cluster-cluster"
0026-05-23 23:39:23 [ ] deploying stack "eksctl-trend-cluster-cluster"
0026-05-23 23:39:53 [ ] waiting for CloudFormation stack "eksctl-trend-cluster-cluster"
0026-05-23 23:40:24 [ ] waiting for CloudFormation stack "eksctl-trend-cluster-cluster"
0026-05-23 23:41:24 [ ] waiting for CloudFormation stack "eksctl-trend-cluster-cluster"
0026-05-23 23:42:24 [ ] waiting for CloudFormation stack "eksctl-trend-cluster-cluster"
0026-05-23 23:43:24 [ ] waiting for CloudFormation stack "eksctl-trend-cluster-cluster"
0026-05-23 23:44:24 [ ] waiting for CloudFormation stack "eksctl-trend-cluster-cluster"
0026-05-23 23:45:25 [ ] waiting for CloudFormation stack "eksctl-trend-cluster-cluster"
0026-05-23 23:46:25 [ ] waiting for CloudFormation stack "eksctl-trend-cluster-cluster"
0026-05-23 23:47:25 [ ] waiting for CloudFormation stack "eksctl-trend-cluster-cluster"
0026-05-23 23:47:26 [!] recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster,
eksctl cannot configure the requested permissions; the recommended way to provide IAM permissions for "vpc-cni" addon i
via pod identity associations; after addon creation is completed, add all recommended policies to the config file, und
r `addon.PodIdentityAssociations`, and run `eksctl update addon`
0026-05-23 23:47:26 [ ] creating addon: vpc-cni
0026-05-23 23:47:27 [ ] successfully created addon: vpc-cni
0026-05-23 23:47:27 [ ] creating addon: kube-proxy
0026-05-23 23:47:28 [ ] successfully created addon: kube-proxy
0026-05-23 23:47:28 [ ] creating addon: coredns
0026-05-23 23:47:29 [ ] successfully created addon: coredns
```